

Mark schemes

Q1.

If candidates have already used $g = 9.81$ do not penalise again on this question.

(a) $0^2 = (28 \sin 50^\circ)^2 + 2 \times (-9.8)s$

M1: Equation to find the max height, with $v = 0$, $u = 28 \sin 50^\circ$ or $u = 28 \cos 50^\circ$ and -9.8 or $-g$

M1

A1: Correct equation

A1

$$s = \frac{(28 \sin 50^\circ)^2}{2 \times 9.8} = 23.5 \text{ m}$$

dM1: Solving for the height

dM1

*A1: Correct height. Awrt 23.5
Note: If using a memorised formula, either 4 marks if final answer correct, 3 marks if substituted correctly but evaluated incorrectly, otherwise zero.*

A1

OR

$$0 = 28 \sin 50^\circ - 9.8t$$

M1: Equation to find time to the max height, with $v = 0$, $u = 28 \sin 50^\circ$ or $u = 28 \cos 50^\circ$ and -9.8 or $-g$

(M1)

$$t = \frac{28 \sin 50^\circ}{9.8} = 2.1887$$

A1: Correct time

(A1)

$$s = 28 \sin 50^\circ \times 2.1887 - 4.9 \times 2.1887^2 = 23.5$$

dM1: Finding the height with their time and $u = 28 \sin 50^\circ$ or $u = 28 \cos 50^\circ$ and -4.9 or $-g/2$

(dM1)

A1: Correct height. Awrt 23.5

(A1)

4

(b) $2 = 28 \sin 50^\circ t - 4.9t^2$

M1: Quadratic equation in t with a
 ± 2 , $u = 28 \sin 50^\circ$ or $u = 28 \cos 50^\circ$
 and -4.9 or $-g/2$.

M1

A1: Correct terms

A1

A1: Correct signs for equation

A1

$$0 = 4.9t^2 - 28 \sin 50^\circ t + 2$$

$$t = 0.0953 \text{ or } t = 4.282$$

dM1: Solving the quadratic equation

dM1

$$t = 4.282 = 4.28 \text{ s (to 3 sf) AG}$$

A1: Correct larger time selected from
 two values.

A1

OR

M1: Calculation of two times, which
 sum or differ to give the time of flight.

(M1)

$$0 = 28 \sin 50^\circ - 9.8t$$

$$t = \frac{28 \sin 50^\circ}{9.8} = 2.1887$$

A1: Correct time by equation for
 zero vertical component of velocity
 or maximum height.

(A1)

OR

$$23.5 = 28 \sin 50^\circ t - 4.9t^2$$

$$t = 2.1887$$

$$21.5 = 4.9t^2$$

dM1: Correct expression for time to fall.

(dM1)

$$t = \sqrt{\frac{21.5}{4.9}} = 2.0947$$

A1: Correct time.

(A1)

$$2.1887 + 2.0947 = 4.2834 = 4.28 \text{ (to 3 sf) } \mathbf{AG}$$

A1: Correct time. Accept 4.29 if their answer rounds to 4.29.

(A1)

5

(c) $v_x = 28 \cos 50^\circ (= 18.00 \text{ m s}^{-1})$

B1: Horizontal component, need not be evaluated.

B1

$$v_y = 28 \sin 50^\circ - 9.8 \times 4.282 = -20.51 \text{ m s}^{-1}$$

M1: Equation for vertical component with $28 \sin 50^\circ$ (or $28 \cos 50^\circ$ if $\sin 50^\circ$ used for horizontal component), -9.8 and awrt 4.28 .

M1

*A1: Correct vertical component.
Awrt ± 20.5*

A1

$$v = \sqrt{18.00^2 + 20.51^2} = 27.3 \text{ m s}^{-1}$$

dM1: Finding speed with a + sign inside the square root.

dM1

A1F: Correct speed. Awrt 27.3.

A1F

5

Intermediate values can be implied by final answer.

[14]

Q2.

$$r = \int v \, dt$$

M1

$$= t^4 + 4 \cos 2t + 5t (+ c)$$

A1

$$\text{When } t = 0, r = 0 \Rightarrow c = -4$$

Finding c correctly

M1

$$\therefore r = t^4 + 4 \cos 2t + 5t - 4$$

A1ft

4

[4]

Q3.

(a) $\mathbf{v} = \frac{dr}{dt}$

M1

$$\mathbf{v} = (e^{\frac{1}{2}t} - 8)\mathbf{i} + (2t - 6)\mathbf{j}$$

i terms

j terms

A1

A1

3

(b) (i) When $t = 3$, $\mathbf{v} = -3.52\mathbf{i}$

Accept $(e^{\frac{3}{2}} - 8)\mathbf{i}$

B1

Speed is $3.52 \, \text{m s}^{-1}$

3.5 does not give 2nd B mark

B1

2

(ii) West

B1

1

(c) $\mathbf{a} = \frac{1}{2}e^{\frac{1}{2}t}\mathbf{i} + 2\mathbf{j}$

M1A1

When $t = 3$, $\mathbf{a} = \frac{1}{2}e^{\frac{3}{2}}\mathbf{i} + 2\mathbf{j}$ or $2.24\mathbf{i} + 2\mathbf{j}$

A1

3

(d) Using $\mathbf{F} = m\mathbf{a}$:

Accept $\mathbf{F} = 7\mathbf{a}$

M1

$\mathbf{F} = 7\left(\frac{1}{2}e^{\frac{3}{2}}\mathbf{i} + 2\mathbf{j}\right)$

$\therefore$ Magnitude of force is $7\left(\left(\frac{1}{2}e^{\frac{3}{2}}\right)^2 + 2^2\right)^{\frac{1}{2}}$

M1

$\mathbf{F} = 21.025$

$\mathbf{F} = 21.0$

Accept 21

A1

3

[12]

Q4.

(a) $13^2 = 0^2 + 2 \times 1.3s$

M1: Use of a constant acceleration equation to find distance.

M1

A1: Correct equation

A1

$$s = \frac{13^2}{2.6} = 65 \text{ m}$$

A1: Correct distance

A1

3

(b) (i) $3900 - 800 - P = 2000 \times 1.3$

M1: Four term equation of motion for car and trailer.

M1

A1: Correct equation

A1

$$P = 3900 - 800 - 2600 = 500 \text{ N}$$

A1: Correct value for P

A1

3

(ii) $T - 500 = 600 \times 1.3$

M1: Three term equation of motion for trailer.

M1

A1: Correct equation

A1F

$$T = 500 + 780 = 1280 \text{ N}$$

A1: Correct tension

A1F

3

[9]

Q5.

(a) $h = \frac{1}{2} \times 9.8 \times 0.6^2 = 1.76 \text{ m AG}$

M1: Equation to find h

M1

A1: Correct value from correct working

A1

2

(b) $x = 18 \times 0.6 = 10.8 \text{ m}$

M1: Calculating range

M1

A1: Correct value

A1

2

(c) $v_y = 9.8 \times 0.6 = 5.88 \text{ m s}^{-1}$

B1: Correct vertical

$$v = \sqrt{5.88^2 + 18^2} = 18.9 \text{ m s}^{-1}$$

M1: Calculating speed

M1

A1: Correct speed

A1

$$\theta = \tan^{-1}\left(\frac{5.88}{18}\right) = 18.1^\circ$$

M1: Calculating angle using tan

M1

A1: Correct angle

A1

18.9 m s⁻¹ at 18.1° below the horizontal.

B1: States below horizontal

B1

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6

[10]

Q6.

(a) $16 = \frac{1}{2}(u + 4.2) \times 5$

M1: Using a constant acceleration equation to find u with v = 4.2 and a = 9.8. Could be derived from a velocity–time graph.

A1: Correct equation

M1A1

$$32 = 5u + 21$$

$$5u = 11$$

$$u = \frac{11}{5} = 2.2 \text{ m s}^{-1}$$

A1: Correct value for u

Eg $s = \frac{1}{2}(u + v)t$ followed by

$16 = (u + 4.2) \times 5$ or similar scores M1A0

A1

OR

First solution from (b) to find acceleration followed by any constant acceleration equation to find u : eg.

(M1)

$$4.2 = u + 0.4 \times 5$$

(A1)

$$u = 2.2$$

(A1)

3

(b) $4.2 = 2.2 + 5a$

M1: Using a constant acceleration equation to find a with $u \neq 0$.

A1F: Correct equation. Follow through for their incorrect u .

M1A1F

$$5a = 2$$

$$a = \frac{2}{5} = 0.4 \text{ m s}^{-2}$$

A1F: Correct value for a , which must be > 0 .

*Follow through for their incorrect u .
(If acceleration found correctly in part (a) and simply quoted as answer to (b) give full marks).*

A1F

OR

$$16 = 2.2 \times 5 + \frac{1}{2} \times a \times 5^2$$

(M1)(A1F)

$$16 = 11 + 12.5a$$

$$a = \frac{5}{12.5} = 0.4 \text{ m s}^{-2}$$

(A1F)

OR

$$16 = 4.2 \times 5 - \frac{1}{2} \times a \times 5^2$$

(M1)(A1F)

$$16 = 21 - 12.5a$$

$$a = \frac{5}{12.5} = 0.4 \text{ m s}^{-2}$$

(A1F)

OR

$$4.2^2 = 2.2^2 + 2a \times 16$$

(M1)(A1F)

$$a = \frac{17.64 - 4.84}{32} = 0.4 \text{ m s}^{-2}$$

(A1F)

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[6]

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Q7.

(a) $20 \sin 50^\circ t - 4.9 t^2 = 0$

M1: Equation to find time, with $y = 0$,

$u = 20 \sin 50^\circ$ or $u = 20 \cos 50^\circ$ and ± 9.8 or $\pm g$.

A1: Correct equation

M1A1

$$t = \frac{20 \sin 50^\circ}{4.9} \text{ or } 3.126 \dots = 3.13 \text{ s} \quad \mathbf{AG}$$

dM1: Solving for t .

A1: Correct time from correct working.

Must see division by 4.9 oe or more than 3 sf

Verification methods can only gain first 2 marks

Special case $t = \frac{15.3}{4.9} = 3.12 \text{ or } 3.13$ scores
M1A1dM1A0

dM1A1

OR

$$0 = 20 \sin 50^\circ - 9.8t$$

$$t = \frac{20 \sin 50^\circ}{9.8} = 1.563$$

$$T = 2 \times 1.563$$

M2: doubling time to max height (could use cos instead of sin) but must use ± 9.8 or $\pm g$.

(M2)

$$= 3.13$$

A2: Correct time from correct working.

Don't penalise use of $g = 9.81$ if already done earlier on script. Would obtain time as 3.12 seconds.

Note: If using a memorised formula either 4 marks if final answer correct, 3 marks if substituted correctly, otherwise zero.

Special case

$$T = 2 \times 1.56 = 3.12 \text{ or } 3.13 \text{ scores M2A1}$$

(A2)

4

(b) $PQ = 20 \cos 50^\circ \times 3.127 = 40.2 \text{ m}$

M1: Calculation of range, could use sin instead of cos.

M1

A1: Correct range
Accept 40.1

A1

2

- (c) No change because a greater mass would not change the acceleration. OR Mass is not used in the equations.

B1: No change

B1

B1: Explanation following a correct statement.

B1

(d) $0 = (20 \sin 50^\circ)^2 + 2 \times (-9.8)s$

*M1: Equation to find height, with
 $u = 20 \sin 50^\circ$ or $u = 20 \cos 50^\circ$ and ± 9.8
or $\pm g$ (and t between 1.56 and 1.57 if
method 2 used).
A1: Correct equation*

M1A1

$$s = \frac{(20 \sin 50^\circ)^2}{2 \times 9.8} = 12.0 \text{ m}$$

*A1: Correct height. Accept 12 or 11.9 or
AWRT 12.0*

A1

OR

$$t = \frac{3.13}{2} = 1.565$$

In method 2, no marks awarded for just finding t .

$$h = 20 \sin 50^\circ \times 1.565 - 4.9 \times 1.565^2$$

(M1)(A1)

$$= 12.0$$

(A1)

*Don't penalise use of $g = 9.81$ if already
done earlier on script. Should still get 12.
Note: If using a memorised formula either
3 marks if final answer correct, 2 marks if
substituted correctly, otherwise zero.*

3

(e) 20 m s^{-1} at 50° below the horizontal.

B1: Speed AWRT 20

B1

*B1: Direction AWRT 50° . Must indicate
below, or down. Could be implied by a
diagram.*

B1

2

[13]

Q8.

(a) $-3 = 70 \sin 10^\circ t - 4.9t^2$

Vertical equation to find t containing a 3

M1

Correct RHS of equation

A1

$$4.9t^2 - 70 \sin 10^\circ t - 3 = 0$$

Correct LHS of equation

A1

$$t = \frac{70 \sin 10^\circ \pm \sqrt{(70 \sin 10^\circ)^2 - 4 \times 4.9 \times (-3)}}{2 \times 4.9}$$

Solving for t

dM1

$$t = -0.226 \text{ or } 2.71$$

Selecting the positive t

dM1

$$t = 2.71$$

Correct time from correct working

A1

(b) $s = 70 \cos 10^\circ \times 2.71 = 187$

Finding horizontal distance

M1

Correct distance

A1

2

(c) $v_x = 70 \cos 10^\circ = 68.9$

Calculating horizontal component of velocity

M1

Correct value

A1

2

- (d) $v_y = 70 \sin 10^\circ - 9.8 \times 2.71 = -14.4$
Calculating vertical component of velocity at $t = 2.71$

M1

Correct value

A1

$$v = \sqrt{68.9^2 + 14.4^2} = 70.4$$

Finding speed

dM1

Correct speed

A1

4

[14]

Q9.

- (a) It is a particle /No air resistance / lift forces act on the ball.

Particle

B1

*Other acceptable assumption
 Deduct one mark for each additional incorrect assumption.*

B1

2

(b)
$$v \sin 40^\circ t - \frac{1}{2} \times 9.8 t^2 = 0$$

Vertical equation to find t .

M1

*Correct equation
 (Equals zero may be implied)*

A1

$$t = \frac{v \sin 40^\circ}{4.9}$$

Solving for t

dM1

Correct t

A1

$$s = V \cos 40^\circ \times \frac{V \sin 40^\circ}{4.9}$$

AG

$$= \frac{V^2 \cos 40^\circ \sin 40^\circ}{4.9}$$

Finding range with their t

M1

*Correct range from correct working
SC Quoting the formula for the range 2 marks.*

A1

6

(c)

$$76 < \frac{V^2 \cos 40^\circ \sin 40^\circ}{4.9} < 82$$

$$\sqrt{\frac{76 \times 4.9}{\cos 40^\circ \sin 40^\circ}} < V < \sqrt{\frac{82 \times 4.9}{\cos 40^\circ \sin 40^\circ}}$$

$$27.5 < V < 28.6$$

An equation to find one value of V .

M1

Correct value for V

A1

Other value of V correct

A1

*Correct range of values
Accept 27.5 – 28.6 but not 28.6-27.5
For using values close to 76 and 82
deduct one mark.*

A1

4

[12]

Q10.

*Use of column vectors is acceptable
throughout this question.*

(a) $\mathbf{v} = 20\mathbf{i} + (-0.4\mathbf{i} + 0.5\mathbf{j})t$

*Use of constant acceleration equation to
find expression for $\mathbf{v}$*

M1

Any correct expression.

A1

2

(b) $\mathbf{v} = (20 - 0.4t)\mathbf{i} + 0.5t\mathbf{j}$

*Simplifying v. (May be implied.)
(Missing brackets may be condoned if followed by correct working.)*

M1

$$20 - 0.4t = 0$$

Putting i component equal to zero

m1

$$t = \frac{20}{0.4} = 50 \text{ seconds}$$

*Correct time
Candidates who are able to see the correct time without supporting working gain full marks.*

Condone $\frac{20\mathbf{i}}{0.4\mathbf{i}} = 50$

A1

3

(c) $\mathbf{r} = 20\mathbf{i} \times t + \frac{1}{2} (-0.4\mathbf{i} + 0.5\mathbf{j}) \times t^2$

Use of constant acceleration equation to find expression for r

M1

Any correct expression

A1

2

(d) (i) $\mathbf{r} = 20\mathbf{i} \times 100 + \frac{1}{2} (-0.4\mathbf{i} + 0.5\mathbf{j}) \times 100^2$

Substituting $t = 100$ into their expression for r (dependent on M1 in part (c))

m1

$$= 2000\mathbf{i} - 2000\mathbf{i} + 2500\mathbf{j}$$

$$= 2500\mathbf{j}$$

Correct simplified position vector ie 2500j

A1

Therefore due north

*Conclusion that helicopter is due north
provided their position vector is of the
form kj , where $k > 0$*

*Note if integration is used there is no need
to prove that the constant is zero.*

*Note marks for (d) (i) can be awarded if
part (c) scores zero.*

A1

3

(ii) $\mathbf{v} = (20 - 0.4 \times 100)\mathbf{i} + 0.5 \times 100\mathbf{j}$

*Substituting $t = 100$ into their expression
for $\mathbf{v}$ (dependent on M1 in part (a)) or use
of other constant acceleration equation
and their position vector (dependent on
M1 in part (c))*

m1

$= -20\mathbf{i} + 50\mathbf{j}$

Correct simplified velocity

A1

$v = \sqrt{20^2 + 50^2} = 53.9$

Correct speed (Accept $10\sqrt{29}$)

A1

*Note marks for (d) (ii) can be awarded if
part (a) scores zero.*

EXAM PAPERS PRACTICE

3

[13]

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Q11.

(a) $6 = 20 \sin 60^\circ t - 4.9t^2$

Vertical equation including a 6

M1

Correct terms

A1

$4.9t^2 - 20 \sin 60^\circ t + 6 = 0$

Correct signs

A1

$$t = \frac{20 \sin 60^\circ \pm \sqrt{(20 \sin 60^\circ)^2 - 4 \times 4.9 \times 6}}{2 \times 4.9}$$

Solving quadratic equation

dM1

$$t = 0.389 \text{ or } t = 3.15$$

$$t = 0.389 \text{ s}$$

Correct solution selected from correct working; AG

A1

5

(b) $x = 20 \cos 60^\circ \times 0.389 = 3.89 \text{ m}$

Finding distance. Correct distance

M1A1

2

(c) $v_x = 20 \cos 60^\circ = 10$

Correct horizontal component

B1

$$v_y = 20 \sin 60^\circ - 9.8 \times 0.389 = 13.5$$

Finding vertical component of velocity

M1

Correct vertical component

A1

$$v = \sqrt{10^2 + 13.5^2} = 16.8 \text{ m s}^{-1}$$

Finding magnitude. Correct magnitude

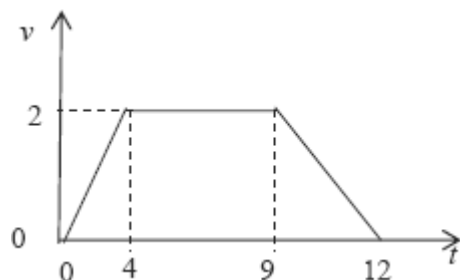
dM1A1

5

[12]

Q12.

(a)



Starts and finishes at rest

B1

Correct shape

B1

Correct values on t-axis

B1

Correct values on v-axis

B1

Condone omission of the origin

4

(b) $s = \frac{1}{2} (5 + 12) \times 2$
or $s = \frac{1}{2} \times 2 \times 4 + 5 \times 2 + \frac{1}{2} \times 2 \times 3 = 17$

Use of the area under the graph (or equivalent) to find s

M1

= 17

*Correct distance
 SC When 21 used instead of 12 allow full marks for s = 26*

A1

2

(c) $\max a = \frac{2}{4} = 0.5$ 25 Exam Papers Practice. All rights reserved.
Maximum acceleration

B1

$300 \times 0.5 = T - 300 \times 9.8$

Three term equation of motion using their a

M1

Correct equation using a = 0.5

A1

$T = 2940 + 150 = 3090$

Correct tension

A1

Q13.

(a) $0^2 = (50 \sin 40^\circ)^2 + 2 \times (-9.8)h$

Equation for h with $v = 0$ and a component of velocity. Correct equation

M1A1

$$h = \frac{(50 \sin 40^\circ)^2}{2 \times 9.8} = 52.7$$

Solving for h

dM1

Correct h

A1

Alt

$$0 = 50 \sin 40^\circ - 9.8t$$

Equation for t with $v = 0$ and a component of velocity

(M1)

$$t = \frac{50 \sin 40^\circ}{9.8} = 3.280$$

Correct t

(A1)

$$h = 50 \sin 40^\circ \times 3.280 - \frac{1}{2} \times 9.8 \times 3.280^2$$

Expression for h with a component of velocity

(dM1)

$$= 52.7$$

ALLOW 52.6

Correct h

(A1)

(b) $6 = 50 \sin 40^\circ t - 4.9t^2$

Forming a quadratic in t . Correct terms

with any signs

M1A1

$$0 = 4.9t^2 - 50 \sin 40^\circ t + 6$$

Correct equation

A1

$$t = \frac{50 \sin 40^\circ \pm \sqrt{(50 \sin 40^\circ)^2 - 4 \times 4.9 \times 6}}{2 \times 4.9}$$

Solving quadratic

dM1

$$= 0.192 \text{ or } 6.37$$

$$t = 6.37$$

Correct solution selected

A2

Alt

$$46.7 = 4.9 t_1^2$$

Finding two times

(M1)

Equation for time to go down

(dM1)

$$t_1 = 3.087$$

Correct time

(A1)

$$t_2 = 3.280$$

Time to go up

(A1)

$$t = 3.087 + 3.280 = 6.37$$

Correct total

(A2)

6

[10]

Q14.

(a) $\mathbf{v} = 1.2\mathbf{i} + (-0.9 \sin t)\mathbf{j}$

Convincingly differentiation

M1A1
2

(b) (i) Speed = $\sqrt{1.2^2 + 0.9^2 \sin^2 t}$

M1A1
2

(ii) Maximum speed = $\sqrt{1.2^2 + 0.9^2}$
Use of $\sin t = 1$

m1

= 1.5

A1
2

[6]

Q15.

- (a) A particle or no spin
First assumption

B1

No air resistance or no wind or only gravity acting
Second assumption

B1

*If more than 2 assumptions given, subtract
 one mark for each incorrect additional
 assumption*

2

- (b) $0 = 25 \sin 40^\circ t - 4.9t^2$
Equation for time of flight

M1

Correct equation

A1

$0 = t (25 \sin 40^\circ - 4.9t)$
Solving for t

dM1

$t = 0 \text{ or } t = \frac{25 \sin 40^\circ}{4.9}$
 Time of flight = 3.28 s

*AG Correct final answer from correct working
 (Verification method M1A1M1A0)*

A1
4

(c) $s = 3.28 \times 25 \cos 40^\circ = 62.8 \text{ m}$

Finding range

M1

Correct range

A1
2

(d) 25 m s^{-1} at 40° **below** the horizontal

Speed

B1

Direction

B1
2

(e) $v_{\min} = 25 \cos 40^\circ = 19.2 \text{ m s}^{-1}$

OR

$$v_{\min} = \frac{62.807}{3.2795} = 19.2 \text{ m s}^{-1}$$

Horizontal component of velocity

M1

*Correct speed
 Accept 19.1 m s^{-1}*

A1
2

[12]

Q16.

(a) $\mathbf{u} = 5\mathbf{i}$ or $\begin{bmatrix} 5 \\ 0 \end{bmatrix}$

Correct velocity

B1
1

(b) $\mathbf{v} = 5\mathbf{i} + (-0.2\mathbf{i} + 0.25\mathbf{j})t$

*Use of constant acceleration equation,
 with $\mathbf{u}$ and $\mathbf{a}$ not zero*

M1

Correct velocity
M1A0 for using 5j or just 5

A1

OR

$$\mathbf{v} = \begin{bmatrix} 5 - 0.2t \\ 0.25t \end{bmatrix}$$

2

(c) $5 - 0.2t = 0$

Easterly component zero

M1

Correct equation

A1

$$t = \frac{5}{0.2} = 25 \text{ seconds}$$

Correct t

A1

3

(d) $\mathbf{r} = 5\mathbf{i} \times 25 + \frac{1}{2}(-0.2\mathbf{i} + 0.25\mathbf{j}) \times 25^2$

Use of constant acceleration equation with
 t from part (c)

M1

Correct expression based on t from part (c)

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$$= 62.5\mathbf{i} + 78.125\mathbf{j}$$

Correct simplification CAO

A1

$$\theta = \tan^{-1}\left(\frac{62.5}{78.125}\right)$$

Using $\tan$ to find the angle

dM1

Correct expression based on t from part (c),
with correct two values (either way)

A1F

$$= 038.7^\circ$$

Correct angle
 Accept 38.6° or 039°

A1

OR

$$\mathbf{r} = \frac{1}{2} (5\mathbf{i} + 6.25\mathbf{j}) \times 25$$

(M1)(A1F)(A1)

$$\theta = \tan^{-1}\left(\frac{5}{6.25}\right) = 038.7^\circ$$

(dM1)(A1F)(A1)

6

[12]

Q17.

(a) $s = ut + \frac{1}{2} at^2$

$$0 = 2\frac{1}{2} ut - \frac{1}{2} gt^2$$

M1

Full method required for time
 (equations of motion, or standard result)

A1

$$0 = t \left(2\frac{1}{2}u - \frac{1}{2}gt \right)$$

m1

$$t = \frac{5u}{g}$$

(if $g = 9.8$ used, lose last A1)

A1

4

(b) $0A = 6u \times \frac{5u}{g}$

M1

$$= \frac{30u^2}{g}$$

CAO

A1

2

(c) $\text{speed}^2 = (6u)^2 + \left(2\frac{1}{2}u\right)^2$

M1

$$\text{speed} = 6\frac{1}{2}u$$

CAO

A1

2

(d) Least speed, at top, = $6u$

B1

1

[9]

Q18.

(a) $\mathbf{v} = 4\mathbf{i} + (-3\mathbf{i} + 12\mathbf{j})t$

M1

use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$

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A1

2

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(b) $t = 0.5, \mathbf{v} = 2.5\mathbf{i} + 6\mathbf{j}$

Ft 2 terms and t subs

B1ft

$$\text{Speed} = \sqrt{2.5^2 + 6^2}$$

2 terms

M1

$$\text{Speed} = 6.5 \text{ ms}^{-1}$$

ft 2 terms

A1ft

3

[5]

Q19.

(a) $v = u + at$

$$0 = 10 + (-0.8) \times t$$

Full method with u, v used correctly, Accept

M1

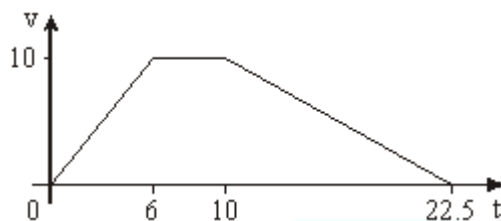
$$t = 12.5 \text{ sec}$$

CAO, (correct substitution and answer)

A1

2

(b)



each line, straight and correct end points

B1 B1 B1

SC first error in times lose B1, subsequent repeated errors, no further penalty

SC B1 for 3 lines giving correct shape but no values shown axes labelled, v, t

B1

4

(c) $\text{distance} = \frac{1}{2} \times 10 \times (4 + 22.5)$

full correct method (including addition of distances)

M1

correct subs, ft graph if final t = 12.5

A1

$$= 132.5 \text{ metres}$$

ft one slip AWRT 133

A1F

3

[9]

Q20.

(a) $\mathbf{d} = 3\mathbf{i} - 6\mathbf{j}$

Accept $\pm \mathbf{d}$ or displacement of 3, 6 shown on a diagram

B1

$$3\mathbf{i} - 6\mathbf{j} = (\mathbf{i} - 2\mathbf{j})t$$

Or equivalent method for t

Accept ratio of vectors leading directly to ± 3

M1

$$t = 3$$

CAO

A1

3

(b) (i) $\mathbf{r} = (\mathbf{i} - 2\mathbf{j}) \times 4 + \frac{1}{2} \times 2\mathbf{j} \times 16$

M1

*full method for vector expression giving change in position
for correct subs (give $4\mathbf{i} + 8\mathbf{j}$)*

A1

$$+6\mathbf{i} - 4\mathbf{j}$$

M1

$$= 10\mathbf{i} + 4\mathbf{j}$$

*ft slip provided obtain vector expression
($\mathbf{u} = 0$ gives $6\mathbf{i} + 12\mathbf{j}$)*

A1F

4

(ii) $A(3,2) \ C(10,4)$

$$\mathbf{d} = 7\mathbf{i} + 2\mathbf{j}$$

*Attempt to find vector **AC** or **CA**
(using candidate's C)*

M1

$$|\mathbf{d}| = \sqrt{7^2 + 2^2}$$

$$AC = \sqrt{53} = 7.28$$

ft d provided two non-zero components

Accept $\sqrt{53}$

A1F

2

[9]

Q21.

(a) $57 = 24 \cos 40^\circ \times t$

component attempted and acceleration = 0

M1

all correct

A1

$$t = 3.10 \text{ sec}$$

CAO

A1

3

(b) $h = 24 \sin 40^\circ \times 3.1 - \frac{1}{2} \times 9.8 \times 3.1^2$

component attempted & acceleration = 9.8

M1

all correct

A1

$$h = 0.734 \text{ m}$$

ft one slip e.g. + 9.8 used

Accept 2 SF answer, AWRT 0.71-0.74

A1F

3

(c) (i) horizontal, $u = 24 \cos 40^\circ = 18.39 \text{ ms}^{-1}$

Seen anywhere in (c), accept 18.4

B1

vertical, $v = 24 \sin 40^\circ - 9.8 \times 3.1$

component attempted & acceleration = 9.8

M1

$$v = -14.95 \text{ ms}^{-1}$$

(accept -15.0)

A1

$$V = \sqrt{(18.39^2 + (-14.95)^2)}$$

Use of candidate's u and 'new' v

M1

$$V = 23.7 \text{ ms}^{-1}$$

ft use of candidate's u and v

A1F

5

(ii) $\tan \theta = \frac{14.95}{18.39}$

Use of candidate's u and v (new)

Accept inverted ratio

M1

$$\theta = 39.1^\circ \text{ or } 39.2^\circ, 140.8^\circ, 140.9^\circ \text{ accept } \pm$$

ft use of candidate's u and v and V

A1

2

[13]

Q22.

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(a) $s = 0 + \frac{1}{2} \times 9.8 \times 4^2$

Full method

M1

Correct subs, accept ± 9.8

A1

$$s = 78.4 \text{ metres}$$

CAO (need positive)

A1

3

- (b) Average speed $= \frac{78.4}{4}$
Also accept full method with use of velocities at $t = 0$ and 4, or at $t = 2$

M1

$$= 19.6 \text{ ms}^{-1}$$

FT distance

A1F

2

- (c) Only force acting is weight
*Acc resistance forces negligible or ignored,
 (not friction, or air friction)*

B1

1

[6]

Q23.

- (a) $v = u + at$

$$0 = 10 + (-0.8) \times t$$

Full method with u , v used correctly

Accept ± 0.8

M1

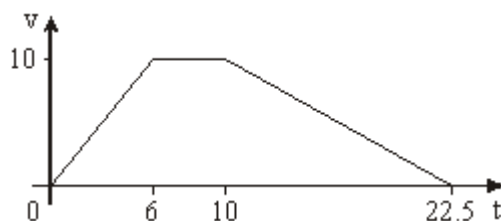
$$t = 12.5 \text{ sec}$$

CAO (correct subs and answer)

A1

2

- (b)



each line, straight and correct end points

B1 B1 B1

*SC: B1 for 3 lines giving correct shape
 but no values shown*

SC: first error in labelling times loses B1,

repeated errors no further penalty
axes labelled v , t

B1

4

(c) $\text{distance} = \frac{1}{2} \times 10 \times (4 + 22.5)$

Full correct method

M1

Correct subs, FT graph if final $t = 12.5$

A1F

= 132.5 metres

FT one slip, AWRT 133

A1F

3

- (d) Acceleration unlikely to:
change so abruptly or be constant
or velocity unlikely to be constant

B1

1

[10]

Q24.

(a) $\mathbf{v} = (6t^2 - 2t)\mathbf{i} + (1 - 12t^2)\mathbf{j}$

differentiating both components

M1

one component correct

A1

second component correct

A1

3

(b) (i) $\mathbf{v}\left(\frac{1}{3}\right) = \left(\frac{6}{9} - \frac{2}{3}\right)\mathbf{i} + \left(1 - \frac{12}{9}\right)\mathbf{j} = -\frac{1}{3}\mathbf{j}$

substituting the value for t into their $\mathbf{v}$

M1

correct velocity

A1

2

(ii) Travelling due south

correct description (Follow through from $\mathbf{v} = \pm k\mathbf{j}$)

A1ft

1

(c) $\mathbf{a} = (12t - 2)\mathbf{i} - 24t\mathbf{j}$

differentiating their velocity

M1

$\mathbf{a}(4) = 46\mathbf{i} - 96\mathbf{j}$

correct acceleration at time t

A1

correct acceleration at $t = 4$

A1

3

(d) $\mathbf{F} = 6(46\mathbf{i} - 96\mathbf{j}) = 276\mathbf{i} - 576\mathbf{j}$

apply Newton's second law correctly

M1

$F = \sqrt{276^2 + 576^2} = 639 \text{ N}$

finding magnitude

M1

correct magnitude

A1

3

or

$a = \sqrt{46^2 + 96^2} = 106.45$

$F = 6 \times 106.45 = 639 \text{ N}$

[12]

Q25.

(a) (i) $40 = 12 + 100a$

Use of a constant acceleration equation to form equation for a

M1

$a = \frac{40 - 12}{100} = 0.28 \text{ ms}^{-2} \quad \text{AG}$

AG; correct answer from correct working

A1

2

(ii) $s = \frac{1}{2}(12 + 40) \times 100$

Expression for distance, using $t = 100$

M1

$= 2600\text{m}$

Correct final distance

A1

2

(c) $F - 40000 = 200 \times 1000 \times 0.28$

Three term equation of motion

M1

$F = 40000 + 56000 = 96000 \text{ N}$

Correct equation

A1

Correct force

A1

3

[7]

Q26.

(a) (i) $30 \sin 60^\circ t - 4.9t^2 = 0$

$t = \frac{30 \sin 60^\circ}{4.9} = 5.302 \quad \text{AG}$

Expression for height equal to zero

M1

Correct expression

A1

AG; correct conclusion from correct working

A1

3

(ii) $s = 30 \cos 60^\circ \times 5.302 = 79.5 \text{ m}$

Use of horizontal component of velocity to find the range

M1

Correct range

A1

2

(b) $30 \sin 60^\circ t - 4.9t^2 = 5$
 $4.9t^2 - 30 \sin 60^\circ t + 5 = 0$
 $t = 0.200$ or 5.102
 $s = 30 \cos 60^\circ \times 5.102 = 76.5 \text{ m}$

Expression for height equal to 5

M1

Correct LHS

A1

Correct RHS

A1

Solving a quadratic equation

m1

Correct solutions

A1

Correct distance

A1

6

[11]

Q27.

(a) $-\mathbf{i} + \mathbf{j} = 2\mathbf{i} - \mathbf{j} + 10\mathbf{a}$

Use of velocity equation

M1

$\mathbf{a} = -0.3\mathbf{i} + 0.2\mathbf{j}$

Correct equation

A1

Correct a

A1

3

(b) $\mathbf{r} = (2\mathbf{i} - \mathbf{j})t + \frac{1}{2}(-0.3\mathbf{i} + 0.2\mathbf{j})t^2 + 20\mathbf{i}$

Use of constant acceleration equation for position

M1

$$= (2t - 0.15t^2 + 20)\mathbf{i} + (-t + 0.1t^2)\mathbf{j}$$

Correct i component

A1

Correct j component

A1ft

3

ft incorrect acceleration

(c) (i) $\mathbf{r}(20) = (2 \times 20 - 0.15 \times 20^2 + 20)\mathbf{i} + (-20 + 0.1 \times 20^2)\mathbf{j}$
 $= 0\mathbf{i} + 20\mathbf{j}$

Substituting $t = 20$ into their expression for r

M1

so due north of origin

Correct conclusion from correct working

A1

2

(c) (ii) $\mathbf{v}(20) = 2\mathbf{i} - \mathbf{j} + 20(-0.3\mathbf{i} + 0.2\mathbf{j}) = -4\mathbf{i} + 3\mathbf{j}$

$$v(20) = \sqrt{4^2 + 3^2} = 5 \text{ ms}^{-1}$$

Finding velocity at $t = 20$

M1

Correct velocity

A1

Finding magnitude

m1

Correct speed

A1ft

4

ft incorrect acceleration

[12]

Q28.

(a) Ball is a particle

One appropriate assumption

B1

No air resistance

Second appropriate assumption

B1

(b) (i)

$$0 = 12 \sin 40^\circ - 9.8t$$

$$t = \frac{12 \sin 40^\circ}{9.8} = 0.787 \text{ s}$$

Equation to find time at maximum height

M1

Correct equation

A1

Solving for t

M1

Correct time

A1

4

(ii) $h = 12 \sin 40^\circ \times 0.7871 - 4.9 \times 0.7871^2$
 $= 3.04 \text{ m}$

Substituting time from previous into expression for height

M1

Correct expression

A1

AG; correct height from correct working

A1

3

(c) $2.44 = 12 \sin 40^\circ t - 4.9t^2$

$$4.9t^2 - 12 \sin 40^\circ t + 2.44 = 0$$

$$t = 0.4385 \text{ or } 1.136$$

$$s = 12 \cos 40^\circ \times 1.136 = 10.4 \text{ m}$$

Equation for time to get to the bar, based on height being 2.44

M1

Correct LHS

A1

Correct RHS

A1

Solving quadratic

m1

Correct time / times

A1

Substituting their larger time into an expression for the horizontal displacement

M1

Correct distance

A1

7

[16]

Q29.

- (a) Ball is a particle/no spin

B1: One assumption

B1

No air resistance/Only gravity or weight

B1: Second assumption

B1

2

- (b) (i) $24.5t - 4.9t^2 = 0$

M1: Equation for vertical motion with height zero

M1

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AG

A1: Correct equation

A1

dM1: Solving for t

dM1

A1: Correct time from correct working

A1

4

- (b) (ii) $R = 10 \times 5 = 50 \text{ m}$

M1: Use of horizontal component of velocity to find the range

M1

A1: Correct range

A1
2

(c) $20 = 10 t$

M1: Horizontal equation

M1

$t = 2$

A1: Time to reach wall

A1

$h = 24.5 \times 2 - 4.9 \times 2^2 = 29.4 \text{ m}$

*dM1: Vertical equation for height with
 $u = 24.5$ and a negative acceleration*

dM1

A1: Correct height

A1
4

- (d) No change as acceleration and initial velocity do not change with the mass

B1: No change

B1

B1: Explanation

B1
2

[14]

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Q30.

(a) $20 = \frac{1}{2} (3 + V)4$

Full method, ($a = 1$)

M1

$V = 7$

A1
2

(b) (i) $S = 2V$ (=14)

Alternative: Time = $T - 4$

M1

$$35 - S = \frac{1}{2} \times t \times V$$

Full method $35 = \frac{1}{2}(T - 4 + 2) \times V$

M1

$$21 = \frac{7}{2}t_2$$

Correct subs $35 = \frac{1}{2}(T - 2)7$

A1

$$t = 6 \quad t = 12$$

ft one slip $T = 12$

A1F

4

(ii) $f = \pm \frac{7}{6} = 1.17 \quad (1.1666)$
Accept $\pm$ ft time

M1A1F

2

(c) Average speed = $\frac{\text{Total distance}}{\text{time}} = \frac{55}{12}$

Attempt at full area for distance

M1

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$$= 4 \frac{7}{12} \quad (4.5833)$$

ft time

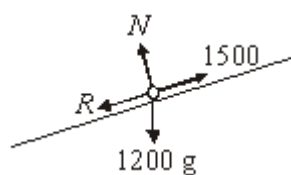
A1F

2

[10]

Q31.

(a) (i)



3 correct & labelled

B1

All correct & labelled, no extras

B1

2

Ignore additional components of weight

(ii) $1500 = R + 1200 \times 9.8 \times \frac{1}{14}$

M1: 3 terms with component attempted
 ($\alpha = 4.096$) – 1 each error

M1A2

$R = 660$

AWRT

A1

4

(iii) 15 ms^{-1}

B1

1

(b) (i) $\frac{1500}{-1200}$

Accept $\pm$

M1

Mag of retardation = 1.25 ms^{-2}

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A1

2

(ii) $0 = 15 + (-1.25) t$

velocities correct, but accept $\pm$ acceleration

M1

$t = 27.3 \text{ sec}$ (27.27)

ft one slip

A1F

2

[11]

Q32.

(a) $0 = (14 \cos 60)^2 - 2 \times 9.8 \times h$

Full method and component of u attempted, v may be present

M1

-1 each error including v ≠ 0

A2

$h = 2.5 \text{ m}$

AWRT ft cos 30, or +g used

A1F

4

(b) $2.5 = \frac{1}{2} \times 9.8 \times t^2$

*M1: full method for time or half time of flight
ft h, need signs consistent for A1*

M1A1F

$t = 0.714 \text{ sec} \left(\text{or } \frac{5}{7} \right)$

A1F

3

[7]

Q33.

(a) (i) $V = 0 + 3 \times 0.8$

M1

$V = 2.4$

AG

A1

2

(ii) $S = \frac{1}{2} (5 + 8) \times 2.4$

Method for total area

M1

A1

$S = 15.6 \text{ m}$

ft slip

A1F

3

(b) $\frac{1}{2} \times T \times 2.4 = 4.8$

Full method for time

M1

$T = 4$

A1

Average speed $= \frac{15.6 + 4.8}{8 + 4}$
Total distance / total time

M1

$= 1.7 \text{ ms}^{-1}$

ft one error in S or in T

A1F

4

[9]

Q34.

(a) $v^2 = 5^2 + 2 \times (-1.8) \times 2.5$

3 terms for M1, accept 1.8; A1 all correct

M1A1

$v = \pm 4 \text{ ms}^{-1}$

Both required

A1

3

(b) $v = u + at, \quad -4 = 4 - 1.8t$

M1 for full method for finding the two times at B or their difference

A1F if one positive & one negative time

M1A1F

$t = 40/9 \text{ or } t = 4 \frac{4}{9} \text{ or } 4.44 \text{ sec}$

A1F for completion, including difference of times

A1F

Alternatives:

$$s = ut + \frac{1}{2} at^2, 2.5 = 5t - \frac{1}{2} \times 1.8t^2$$

(M1)

$$(9t - 5)(t - 5)$$

(A1)

$$\text{time difference} = 4 \frac{4}{9}$$

(A1F)

If time from B to stopping point found,

20/9, M1A1F

time $\times 2$, 40/9, A1F

[6]

Q35.

(a) (i) $a = \frac{(5U - 3U)}{(5 - 2)}$
full method for a

M1

$$a = \frac{2U}{3}$$

Accept 0.67U

A1

2

(ii) $s = \frac{1}{2} (5U + 3U) \times 3$

full method for s or attempt at area of whole shape

M1

$$s = 12U$$

ft incorrect a used

A1F

2

(b) $20 = \frac{1}{2} (5U + 0) \times 4$

full method, use of equation or area of triangle
($a = -1.25 U$ if used)

M1A1

$$U = 2$$

ft slip in line above e.g. miscopy

A1F

3

Alternative: average speed = 5

M1

equation for U

A1

$$U = 2$$

A1

[7]



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